

Weak lensing of the DESI DR1 Lyman- α forest by low-redshift galaxies and quasars: estimator, validation and a first amplitude constraint

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Abstract

Gravitational lensing by structure at $z < 2$ displaces the Lyman- α forest sightlines coherently, which dilates and shears the transverse correlation of the forest flux. We measure this effect with a pair-based quadratic estimator: the products of forest pixels on pairs of sightlines are weighted by the response of the measured forest correlation to a deflection and projected on a template of the lensing deflection built from DESI DR1 galaxies and quasars (BGS, LRG, ELG and QSO samples) and BOSS DR12 galaxies at $0.1 < z < 1.6$ in five redshift slices. The tracer template shares no density modes with the forest at $2.1 < z < 3$, so unlike a CMB-lensing template it is free of the intrinsic forest–density three-point term that we found to bias the CMB cross-correlation at the 0.4 level of the lensing amplitude. On 400 self-lensed mock realisations with lognormal tracers the estimator is unbiased (null -0.05 ± 0.08 , recovery 0.91 ± 0.08 , paired response 0.966 ± 0.025 for a true amplitude of one). The forest correlation that sets the response kernel is fitted with explicit redshift evolution ($b_F \propto (1+z)^{3.5}$), which accounts for half of the amplitude difference between pixel pairs in the Ly α and Ly β rest-frame windows. The tracer biases are measured from NaMaster auto-spectra and the deflection templates themselves are validated against ACT DR6 and Planck PR4 CMB lensing: relative to the prediction for the template, the deflection cross-spectrum has amplitude 0.99 ± 0.04 (ACT) and 0.92 ± 0.03 (Planck). Applied to every DR1 forest pixel below $z = 3$ (427 888 sightlines with $1.96 \leq z \leq 3.0$, effective redshift $z_{\text{eff}} = 2.35$, over $11\,000\,\text{deg}^2$, both delta regions), the forest auto-correlation gives a lensing amplitude relative to the Λ CDM prediction of $A = 0.35 \pm 0.47$ (single slab; Fisher scale 0.38), with curl, random-template and injection tests consistent with no systematic. The same estimator applied to the quasar–forest cross-correlation, with its own fitted correlation model (quasar bias, redshift offset and smoothing), gives an independent $A = 0.61 \pm 0.52$, and the optimal combination is $A = 0.47 \pm 0.35$, i.e. $A < 1.04$ at 95 per cent; with the template bands widened to $L = 1000$, where the templates are still validated against CMB lensing, the combination is $A = 0.58 \pm 0.31$. A detection at this depth was not expected; the full DESI sample should reach $2\text{--}3\sigma$.

1 Summary

This note documents the measurement in a self-contained way: the data (Section 2), the estimator and its templates (Section 3), the mock validation that defines its acceptance (Section 4), the DR1 result with its null tests (Section 5) and the caveats (Section 7). The derivations of the estimator and the CMB-lensing forecasts are in the project’s main report (`report/main.pdf`); the pipeline is `code/pipeline` (mocks, estimator, campaign) and `code/stageb` (DR1). Every number quoted here is in the committed JSON products under `report/`.

2 Data

Forest. The DESI DR1 Lyman- α forest deltas [DESI Collaboration et al., 2026, Ramírez-Pérez et al., 2024], extracted with `picca` v9 on a $0.8\,\text{\AA}$ grid with the lines, DLA (combined CNN +

Gaussian-process catalogue, $N_{\text{HI}} > 10^{20.3} \text{ cm}^{-2}$) and BAL masks applied by the DR1 extraction. The catalogue provides two rest-frame windows and we use both: region A, 1040–1205 Å, the Ly α forest proper, and region B, 920–1020 Å, blueward of the Ly β emission line. A region-B pixel at observed wavelength λ carries Ly α absorption at $z = \lambda/1215.67 - 1$, which is the field we want, and simultaneously Ly β absorption at $z = \lambda/1025.72 - 1$; over our slab the second sits at $z = 2.7$ – 3.7 , more than $1000 h^{-1} \text{ Mpc}$ further away. A pixel pair inside $30 h^{-1} \text{ Mpc}$ that joins an A and a B pixel therefore measures Ly α –Ly α alone, while a B–B pair also carries Ly β –Ly β at a common redshift; we use A $\times$ A and A $\times$ B pairs and drop B $\times$ B. Region B is treated as an extension of the same sightline: the two segments are concatenated and ordered in distance, each pixel carrying a region tag, and each segment keeps its own continuum-fit block (Section 3.2).

We keep every pixel of the DR1 grid below $z = 3.0$: the grid starts at 3600 Å, i.e. $z = 1.96$, in both windows, so the slab is $1.96 \leq z \leq 3.0$ (the first version of this note cut at $z = 2.1$, a limit inherited from the mock box that discarded 16 per cent of the region-A pixel weight and 22 per cent of the region-B one). Per region, forests with at least 50 such pixels are kept: 427 888 sightlines, 2.80×10^8 pixels of which 19.5 per cent are in region B, $11\,000 \text{ deg}^2$. The weighted mean pixel redshift, which we adopt as the effective forest redshift and as the source plane of the templates, is $z_{\text{eff}} = 2.348$ (unweighted mean 2.342, median 2.292; the 2.1–3.0 sample had $z_{\text{eff}} = 2.416$). Pixel weights are the *picca* weights $w = 1/(\eta\sigma_N^2 + \sigma_{\text{LSS}}^2(\lambda))$ with $\eta \simeq 1$ and $\sigma_{\text{LSS}}^2 = 0.07$ – 0.19 : the diagonal of the inverse covariance, including the signal variance. The per-forest noise power $P_N = \sigma_N^2 \Delta\chi_{\text{pix}}$ has median $0.54 h^{-1} \text{ Mpc}$ and a scatter of 2.2 in $\ln P_N$.

Tracers. Three catalogues. (i) The DESI DR1 large-scale-structure clustering catalogues v1.5 [Ross et al., 2025]: LRG ($0.4 < z < 1.1$), ELG_LOPnotqso ($0.8 < z < 1.6$), QSO ($z < 1.75$ here) and the magnitude-limited bright-galaxy sample BGS_BRIGHT–21.5 ($0.1 < z < 0.4$), with their completeness, redshift-failure and imaging-systematics weights, and two random catalogues per Galactic cap (5000 deg^{-2}) carrying redshifts and weights. (ii) The BOSS DR12 combined LOWZ+CMASS galaxy sample [CMASLOWZTOT; Reid et al., 2016], $0.1 < z < 0.8$, with the weight $w_{\text{systot}}(w_{\text{cp}} + w_{\text{noz}} - 1)$ and one random catalogue per cap; BOSS covers sky that DR1 does not (and vice versa), which is why it is included. They are split into five redshift slices, 0.1–0.4 (BGS, BOSS), 0.4–0.6 (LRG, BOSS), 0.6–0.8 (LRG, BOSS), 0.8–1.1 (LRG, ELG, QSO) and 1.1–1.6 (ELG, QSO), ending at $z = 1.6$ so that the tracers stay $400 h^{-1} \text{ Mpc}$ in front of a forest that now starts at $z = 1.96$ (the 1.6–1.75 quasar slice of the first version held 1–2 per cent of the convergence power and is dropped). Figure 1 shows where they sit under the lensing kernel of a source at $z_{\text{eff}} = 2.35$: the five slices hold about 95 per cent of the power of $\kappa_{\text{Ly}\alpha}$, the 0.1–0.4 slice alone 19–29 per cent.

CMB lensing. The ACT DR6 baseline convergence [Madhavacheril et al., 2024, Qu et al., 2024] and the Planck PR4 convergence [the “2018-like” minimum-variance reconstruction of Carron et al., 2022, rotated from Galactic to equatorial coordinates] are used to validate the tracer biases and the deflection templates (Section 3.3).

3 Estimator

3.1 Lensing of pair correlations

Let $\delta_F(\boldsymbol{\theta}, \chi)$ be the forest flux fluctuation observed along a sightline at angular position $\boldsymbol{\theta}$ and comoving distance χ . Lensing maps the observed position to the true one, $\boldsymbol{\theta}_{\text{true}} = \boldsymbol{\theta} + \boldsymbol{\alpha}(\boldsymbol{\theta})$, with $\boldsymbol{\alpha} = \nabla\phi$ and $\kappa = -\nabla^2\phi/2$, so that the observed field is the true field sampled at $\boldsymbol{\theta} + \boldsymbol{\alpha}$. For a pair of sightlines a, b the correlation of two pixels at transverse separation $r_{\perp} = \chi|\boldsymbol{\theta}_a - \boldsymbol{\theta}_b|$ and radial separation $r_{\parallel}$ is the true correlation at the displaced separation,

$$\langle \delta_F(a, \chi_p) \delta_F(b, \chi_q) \rangle = \xi_F(|\boldsymbol{\theta}_{ab} + \boldsymbol{\alpha}_a - \boldsymbol{\alpha}_b|, \chi, r_{\parallel}) \simeq \xi_F(r_{\perp}, r_{\parallel}) + \frac{\partial \xi_F}{\partial r_{\perp}} \chi \hat{\boldsymbol{\theta}}_{ab} \cdot (\boldsymbol{\alpha}_a - \boldsymbol{\alpha}_b), \quad (1)$$

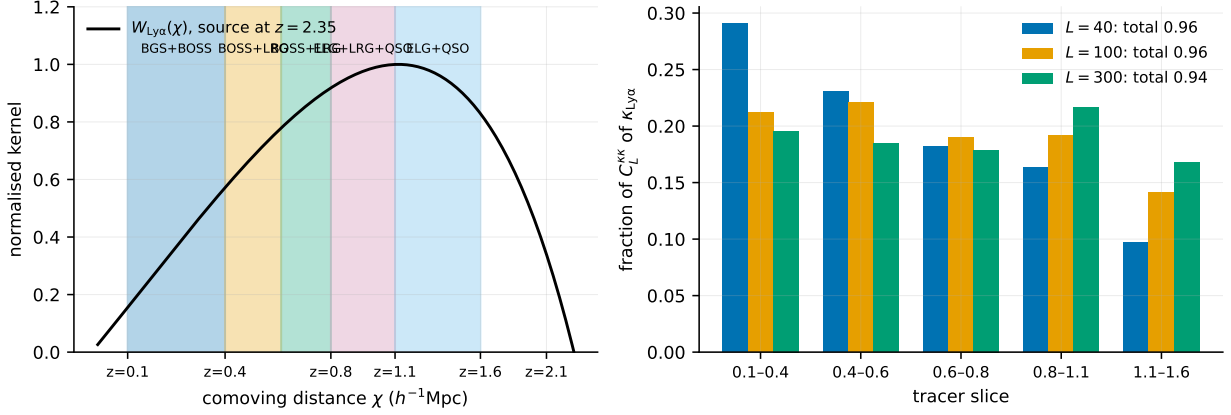


Figure 1: Left: the lensing kernel of a source at $z_{\text{eff}} = 2.35$, the weighted mean pixel redshift of the forest sample, with the five tracer slices shaded and their tracers named. Right: the fraction of the power of $\kappa_{\text{Ly}\alpha}$ contributed by each slice at $L = 40, 100$ and 300 (Limber, non-linear matter power).

with $\hat{\theta}_{ab}$ the unit pair direction (from b to a). For a pair separation small compared with the wavelength of the lens, $\alpha_a - \alpha_b \simeq (\theta_{ab} \cdot \nabla) \nabla \phi$: the convergence dilates the transverse correlation isotropically and the shear makes it anisotropic. Equation (1) is exact in the pair displacement and does not require the squeezed limit; the deflection difference is evaluated from the template at the two sightlines.

3.2 Quadratic estimator with a deflection template

Given a template deflection $\alpha^T(\theta)$ for the lensing field, the amplitude A of the deflection in the data ($\alpha = A \alpha^T$ in expectation) is estimated from all pixel pairs on all sightline pairs with $3 \leq r_\perp \leq 30 h^{-1} \text{Mpc}$ and $|r_\parallel| < 30 h^{-1} \text{Mpc}$,

$$\hat{A} = F^{-1}(q - \bar{q}), \quad q = \sum_{ab} [\chi_{ab} \hat{\theta}_{ab} \cdot (\alpha_a^T - \alpha_b^T)] \sum_{p \in a, q \in b} w_p w_q \delta_F(p) \delta_F(q) \xi'_F(r_\perp, r_\parallel), \quad (2)$$

with $\xi'_F = \partial \xi_F / \partial r_\perp$ the response kernel, $\bar{q}$ the mean field (the same sum with $\delta_F(p) \delta_F(q)$ replaced by ξ_F : it removes the correlation of the sightline density with the template and is not zero on a masked footprint), and F the response matrix, $\sum w_p w_q \xi_F'^2 [\chi \hat{\theta} \cdot \Delta \alpha^T]^2$ up to the same geometric factors. This is the optimal quadratic estimator for the pixel-diagonal inverse covariance; a full per-sightline inverse covariance would shape the weights along $k_\parallel$ and we estimate its gain at 8 per cent in signal-to-noise, not worth its complexity here. The pair geometry uses a local tangent-plane metric per pair, valid at the separations used; the sightline density is $19\text{--}22 \text{ deg}^{-2}$ so the pair count is 5.1×10^6 . A pixel pair is skipped when both of its pixels are in region B.

The template deflection is split into five multipole bands, $40 \leq L < 100$, $100\text{--}200$, $200\text{--}300$, $300\text{--}400$ and $400\text{--}500$ (cosine-tapered windows), plus the complementary “junk” band, and each band has a curl partner (the deflection rotated by 90°). The tracer auto-spectra follow linear theory to $\ell \simeq 600$ (Figure 5) and the template alm reaches $\ell = 1000$, so the two bands above 300 carry signal; the deflection is evaluated at nside 1024 to resolve the top band. The eleven components are fitted jointly with the 11×11 response matrix; the science amplitude is the common amplitude of the three gradient bands with the curl and junk amplitudes marginalised (a “curl null” is reported as the mean curl amplitude). Errors are from a HEALPix jackknife on the pair midpoints (nside 8, 301 regions on DR1) and are checked against the Fisher scale $F^{-1/2}$ and against random templates.

The response kernel. $\xi_F(r_\perp, r_\parallel)$ is measured from the same forests in $1 h^{-1} \text{Mpc}$ cells and fitted with the model

$$\xi_F(r_\perp, r_\parallel) = \xi_{\text{ref}}(r_\perp, r_\parallel; b_F^2, \beta_F) + \sigma(r) S(r_\perp, r_\parallel) + N(r_\perp) \mathbb{I}[|r_\parallel| < 1 h^{-1} \text{Mpc}], \quad (3)$$

where ξ_{ref} is the two-parameter linear model $P_F(\mathbf{k}) = b_F^2(1 + \beta_F \mu^2)^2 P(k)$ pushed through the DESI pixel and resolution windows and the per-forest continuum operator; S is a tensor-product cubic B-spline in $(r_\perp, r_\parallel)$ carried on a strictly positive envelope $\sigma(r)$, $r = (r_\perp^2 + r_\parallel^2)^{1/2}$ (the monopole of ξ_{ref}), which removes the dynamic range of ξ_F from the spline coefficients; and N is a one-dimensional spline confined to the first radial bin. Every basis element is averaged over the $1 h^{-1}\text{Mpc}$ cell exactly as the pair counts are binned, and all 50 parameters are fitted by weighted least squares with the accumulated pair weights, so that ξ_F and its *analytic* derivative ξ'_F follow the data rather than a two-parameter shape. Three points matter.

(i) *Why the shape has to be fitted.* The estimator normalisation is $\langle \hat{A} \rangle = A \langle W g_{\text{used}} g_{\text{true}} \rangle / \langle W g_{\text{used}}^2 \rangle$ with $g = \chi \xi'_F$ and W the pair weight, so an error in the *shape* of the kernel biases the amplitude at first order; it does not cancel. The two-parameter table misses the DR1 measurement by 5–12 per cent at $r_\parallel < 5 h^{-1}\text{Mpc}$, which is where 80 per cent of the information is, and its free three-coefficient version returns $\beta_F = 0.43$ from the μ^2 term against $\beta_F = 1.76$ from the μ^4 term: the Kaiser form itself is rejected. Fitting the correction changes χ^2 from 1880 to 235 over 810 cells and raises the amplitude normalisation by 8 per cent.

(ii) *The continuum projection.* The correlation the pixels see is $P_a C P_b^T$ with P_a the mean-and-slope removal of the continuum fit on forest a ; it is the largest single distortion of the model. Earlier iterations evaluated it for one representative forest spanning the whole slab (1295 pixels, $712 h^{-1}\text{Mpc}$) with uniform weights, while the median picca forest has 689 pixels ($397 h^{-1}\text{Mpc}$); that over-predicted ξ_F by 5–20 per cent, growing with $r_\perp$. It is now the pair-weight average of $P_a C P_b^T$ over sampled pairs of real DR1 forests, with the measured pixel weights and with the projector running over every pixel the continuum fit saw (not only the pixels inside $2.1 < z < 3.0$). The corrected projection removes the mean offset but predicts more $r_\perp$ tilt than the data show, which is why the empirical correction is still needed.

(iii) *The same-wavelength term.* Pixels at the same observed wavelength on different sightlines carry an extra positive correlation of $\simeq 10^{-3}$, flat in $r_\perp$ beyond $6 h^{-1}\text{Mpc}$ and significant at $5\text{--}7\sigma$ per cell, absent from the neighbouring radial bins (Figure 3): a sky-subtraction and calibration residual, not a correlation of the density field. It therefore belongs in the mean field, which subtracts $\langle \delta_F \delta_F \rangle$, but carries no lensing response, and N is excluded from ξ'_F for that reason.

On DR1 the fit gives $b_F^2 = 0.0296$ and $\beta_F = 1.04$ over $3 \leq r_\perp < 30$, $r_\parallel < 30 h^{-1}\text{Mpc}$ (Figure 2); the base parameters are not measurements of the forest bias, since the correction absorbs part of the shape.

(iv) *Redshift evolution, and the two pair types.* $A \times A$ and $A \times B$ pairs do not have the same correlation: fitted separately with one non-evolving model they give $b_F^2 = 0.0302$ against 0.0253, an amplitude ratio of 0.84. The $A \times B$ pairs sit at lower redshift (a region-B pixel is bluer than any region-A pixel of the same quasar), so the natural reading is redshift evolution of the forest correlation. The measured cells are therefore split by the pair mean redshift (seven bins between $z = 1.96$ and 3.0, with the pair-weighted mean redshift of every cell recorded) and one evolving model is fitted to all of them, with $x = (1 + z)/(1 + z_{\text{ref}})$, $z_{\text{ref}} = 2.4$,

$$\xi_F(r_\perp, r_\parallel, z) = b_F^2 \left[\frac{D(z)}{D(z_{\text{ref}})} \right]^2 x^{2\gamma_b} [T_0 + 2\beta_F(z)T_1 + \beta_F(z)^2 T_2] + x^{\gamma_S} \sigma(r) S(r_\perp, r_\parallel) + N(r_\perp) \mathbb{K}[r_\parallel < 1], \quad \beta_F(z) = \beta_F x \quad (4)$$

where $T_{0,1,2}$ are the projected basis tables of the μ^0, μ^2, μ^4 terms of ξ_{ref} (linear power at z_{ref} , D the linear growth factor), S and N are as before, and $(\gamma_b, \gamma_\beta, \gamma_S)$ are the only new parameters; the same-wavelength term is instrumental and is not evolved. The fitted table is layered in comoving distance (21 nodes over $2.05 < z < 3.05$) and the pair kernel interpolates between the two layers bracketing each pair's mean distance. On DR1 the base model alone gives $\gamma_b = 3.58$, $\gamma_\beta = -1.64$, i.e. $b_F \propto (1 + z)^{3.6}$ and a total amplitude exponent $d \ln(b_F^2 D^2) / d \ln(1 + z) = 5.2$; with the correction included, $\gamma_b = 3.49$, $\gamma_\beta = -1.37$, $\gamma_S = 5.0$, and the fit quality per redshift bin ($\chi^2 = 206\text{--}238$ per 810 cells) equals that of the non-evolving fit on the collapsed cells (239), so the power laws describe every bin. Since the correction evolves like the base amplitude, the split between the two is degenerate in that fit and only their sum, the kernel, is determined. Refitting the two pair types with the exponents held fixed gives an $A \times B$ to $A \times A$ amplitude ratio of 0.92 at fixed redshift: evolution

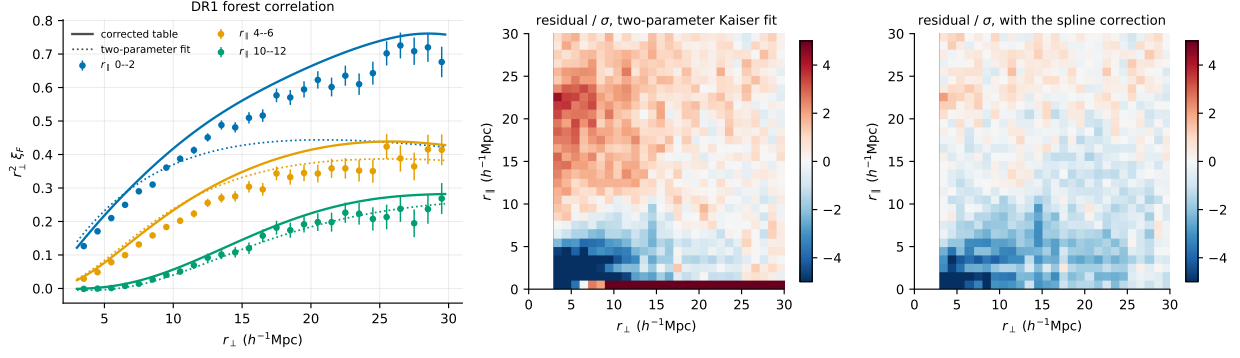


Figure 2: DR1 forest correlation from 1.8×10^8 pixels. Left: measured $r_{\perp}^2 \xi_F$ in three radial-separation strips (points, pair-weight errors) with the fitted table of Equation (3) (solid) and the two-parameter Kaiser fit of the same counts and basis (dotted). Middle and right: the residual of each fit in units of the per-cell error, on the $(r_{\perp}, r_{\parallel})$ plane used by the kernel. The two-parameter fit is systematically high at $1 < r_{\parallel} < 5 h^{-1}\text{Mpc}$, where most of the lensing information is, and low by up to a factor 1.6 in the first radial bin; χ^2 over the 810 fitted cells falls from 1880 to 235.

explains half of the difference, and the remaining 8 per cent is a property of the region-B pairs that the single evolving table does not model (`report/stageb/xi_zev01_dr1.json`). Non-linear Arinyo-i-Prats terms were tested and make the fit *worse* at these separations, monotonically in q_1 ($\chi^2 = 1249, 1300, 1357, 1421$ for $q_1 = 0, 0.3, 0.6, 0.9$ with the old projection), while k_p is degenerate with the pixel window; this is consistent with the DESI DR1 full-shape analysis, which bounds $q_1 < 0.51$ at 95 per cent from the same data and leaves $\{k_v, a_v, b_v, k_p\}$ prior-dominated, and which fits the auto-correlation only at $r > 25 h^{-1}\text{Mpc}$ [Cuceu et al., 2025]. The lensing information is concentrated at small radial separations: 80 per cent of it comes from $r_{\parallel} < 5 h^{-1}\text{Mpc}$ and half from $r_{\perp} < 15 h^{-1}\text{Mpc}$ (Figure 4).

How much freedom to give the correction. The correction is fitted to the same pixel pairs that the estimator multiplies, so a random error δg in the kernel attenuates the amplitude by $\langle W \text{Var}(\delta g) \rangle / \langle W g^2 \rangle$: more parameters buy a better fit and cost response. Both sides are measurable. On DR1 the amplitude gain saturates almost at once — 1.077 with one cubic segment in each direction (16 coefficients for S plus 6 for N), 1.099 with 30, 1.100 with 42 — while the cost, measured on 40 mock realisations in Section 4, does not: the paired response falls by 1.4 ± 1.2 , 4.0 ± 1.8 and 10.4 ± 2.3 per cent for the same three. We therefore adopt the smallest, a bicubic surface in $(r_{\perp}, r_{\parallel}^2)$ with 22 fitted parameters in all, which takes χ^2 from 1880 to 235 over the 810 cells and costs no measurable response.

Robustness. With that choice the fitted kernel is insensitive to the parts of the construction that are choices rather than data. Across four continuum-projection variants (single full-slab forest; pair-weighted with the redshift-cut span; pair-weighted with the full picca span; an independent Monte-Carlo sample of the latter) the amplitude normalisation lies between 1.075 and 1.079 and the Fisher-weighted kernels agree to 2.0–5.3 per cent rms. We adopt the full-picca-span projection and quote 1.077 for the normalisation change, with a 2 per cent systematic covering the knot and projection variations.

3.3 Templates from low-redshift tracers

For a tracer with objects at (θ_i, χ_i) and mean density $\bar{n}(\chi)$ per steradian per unit distance, the kernel-weighted map

$$m(\theta) = \sum_{i \in \text{pixel}} \frac{w_i W(\chi_i)}{b \bar{n}(\chi_i) \Omega_{\text{pix}}} - (\text{randoms, scaled}), \quad W(\chi) = \frac{3}{2} \Omega_m \frac{H_0^2}{c^2} (1+z) \chi \frac{\chi_s - \chi}{\chi_s}, \quad (5)$$

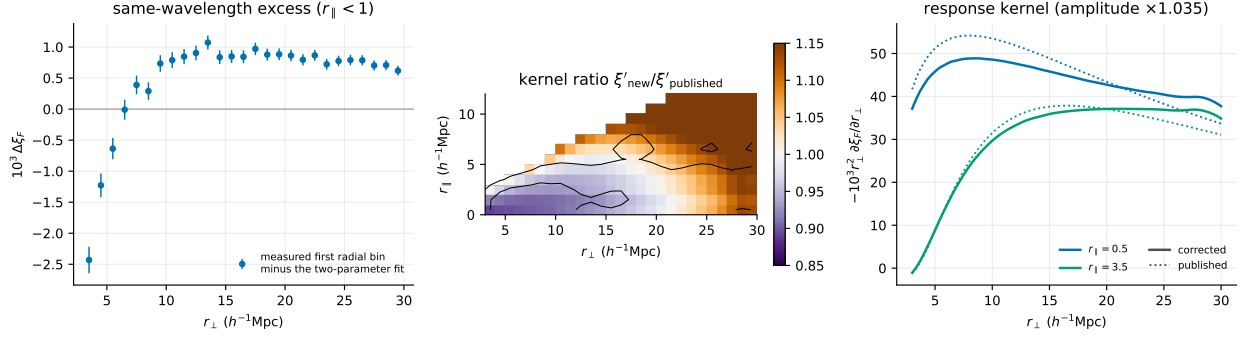


Figure 3: What the correction does. Left: the excess of the first radial bin over the two-parameter fit (points) and the fitted same-wavelength term $N(r_\perp)$ (line); it is flat at $\simeq 10^{-3}$ beyond $6 h^{-1}\text{Mpc}$ and enters the mean field but not the kernel. Middle: the ratio of the new response kernel to the one used in the first measurement, over the part of the pair domain where the kernel is large, with contours of the Fisher-information density of the amplitude. Right: the kernel itself, $-r_\perp^2 \partial \xi_F / \partial r_\perp$, at two radial separations, published (dotted) against corrected (solid). Fisher-weighted, the kernel change raises the lensing amplitude by 10 per cent.

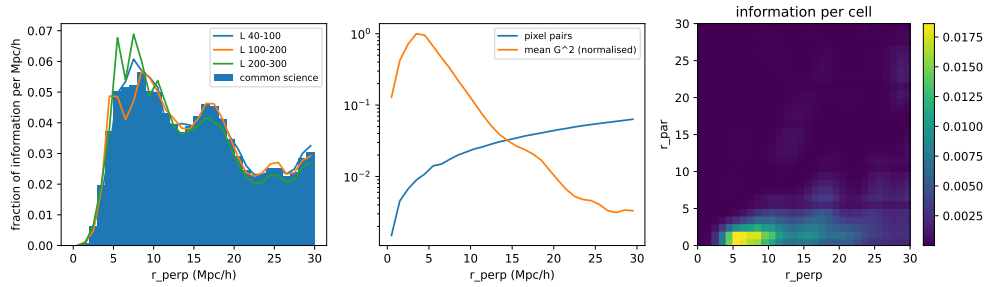


Figure 4: Where the lensing information lives: the cumulative Fisher information of the amplitude versus $r_\perp$ and $r_\parallel$ on a scale-1 mock, per band and combined. Half of the information is inside $r_\perp = 15 h^{-1}\text{Mpc}$ and 80 per cent at $r_\parallel < 5 h^{-1}\text{Mpc}$.

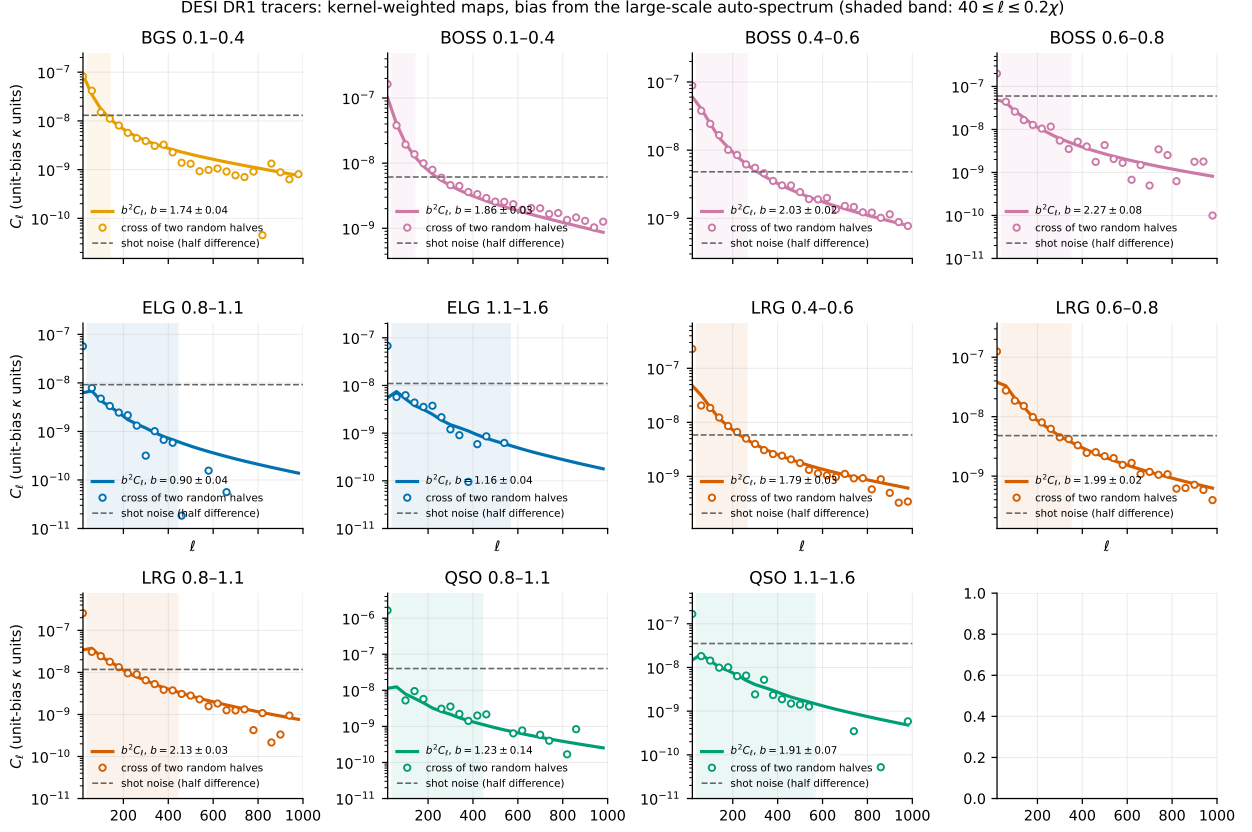


Figure 5: Angular power spectra of the eleven tracer maps in convergence units for unit bias: the NaMaster cross-spectrum of two random halves (points, no shot noise), the fitted $b^2 C_\ell$ through the bandpower windows (line) and the shot noise measured from the half difference (dashed). The shaded band is the fit range $40 \leq \ell \leq 0.2\chi(z_{\text{mid}})$.

estimates the contribution of the tracer’s slice to $\kappa_{\text{Ly}\alpha}$ for a source at $\chi_s = \chi(z_{\text{eff}})$, $z_{\text{eff}} = 2.348$, once the linear bias b of the tracer is known: $\langle m \rangle = \int_{\text{slice}} W \delta d\chi$. The randoms define the footprint (pixels holding at least half the mean random count, which excludes partially covered rim pixels), the completeness (their density smoothed by 1°) and the mean-density subtraction; the map is built at $n_{\text{side}} 512$. The BOSS map is built the same way from its own randoms.

Bias from the angular power spectrum. The bias of every (tracer, slice) is measured from its own map with NaMaster [Alonso et al., 2019]: the cross-spectrum of two random halves of the catalogue (which has no shot noise) is decoupled from the binary mask in bandpowers of width 40 and fitted, over $40 \leq \ell \leq 0.2\chi(z_{\text{mid}})$ ($k \leq 0.2 h \text{ Mpc}^{-1}$) and with the NaMaster Gaussian covariance, against the Limber spectrum of the slice, $C_\ell = b^2 \int_{\text{slice}} W^2 P(k = (\ell + \frac{1}{2})/\chi, \chi) d\chi/\chi^2$, times the pixel window and pushed through the same bandpower windows. This replaces the f_{sky} -scaled pseudo-spectra and the 16-realisation Monte-Carlo mask transfer of the first version (which was 0.73–0.95 over the band for the fragmented DR1 footprint); the biases move by 1–3 per cent for the LRGs and ELGs and by up to 25 per cent for the noisiest quasar slice, where the covariance now weights the bandpowers properly. The shot noise of the full map (needed for the Wiener weights) is the white level of the half-difference map, measured as a pseudo-spectrum over $\langle M^2 \rangle$: white pixel noise is local and that estimate is exact, whereas the NaMaster decoupling of a spectrum that is not band-limited attributes the coupling to $\ell > \ell_{\text{max}}$ to the band and overshoots by 13–16 per cent for the DESI masks (2 per cent for BOSS). It agrees with the weighted Poisson expectation $\sum_i (u_i w_i)^2 \Omega_{\text{pix}}^2 / \Omega_{\text{mask}}$ to 1–3 per cent for the galaxy samples and to 10 per cent for the quasars. Figure 5 shows the spectra and Table 1 the biases.

tracer, slice	objects	b (auto)	b (ACT κ)	ratio	b (Planck κ)	ratio
BGS 0.1–0.4	300 043	1.74 ± 0.04	1.55 ± 0.34	0.89	1.67 ± 0.24	0.96
BOSS 0.1–0.4	403 034	1.86 ± 0.03	1.92 ± 0.31	1.03	1.48 ± 0.22	0.79
LRG 0.4–0.6	506 911	1.79 ± 0.03	1.62 ± 0.22	0.90	1.67 ± 0.19	0.93
BOSS 0.4–0.6	686 370	2.03 ± 0.02	1.58 ± 0.21	0.78	1.74 ± 0.16	0.86
LRG 0.6–0.8	771 894	1.99 ± 0.02	2.09 ± 0.19	1.05	1.68 ± 0.17	0.84
BOSS 0.6–0.8	203 187	2.27 ± 0.08	2.45 ± 0.39	1.08	2.21 ± 0.30	0.98
LRG 0.8–1.1	859 822	2.13 ± 0.03	2.44 ± 0.19	1.15	2.21 ± 0.16	1.04
ELG 0.8–1.1	1 016 365	0.90 ± 0.04	0.59 ± 0.14	0.65	0.57 ± 0.12	0.63
QSO 0.8–1.1	143 846	1.23 ± 0.14	1.02 ± 0.25	0.83	1.68 ± 0.22	1.37
ELG 1.1–1.6	1 415 707	1.16 ± 0.04	1.17 ± 0.14	1.00	1.03 ± 0.12	0.89
QSO 1.1–1.6	358 775	1.91 ± 0.07	2.22 ± 0.20	1.16	1.73 ± 0.18	0.91

Table 1: Linear biases of the eleven tracer slices from their large-scale auto-spectra (NaMaster cross-spectrum of two random halves, $40 \leq \ell \leq 0.2\chi$) and from the cross-spectra with the ACT DR6 and Planck PR4 lensing maps (`report/stageb/cmb_bias_check{, _planck}.json`). The DESI clustering analyses quote $b \simeq 1.9$ –2.3 for the LRGs, 1.3–1.4 for the ELGs, $b_{\text{QSO}}(z) = 0.278[(1+z)^2 - 6.565] + 2.393$ for the quasars and $\simeq 1.5$ for the BGS; BOSS quotes $\simeq 2$ for LOWZ and CMASS.

Cross-check with CMB lensing. The NaMaster cross-spectrum of each unit-bias map with the ACT DR6 and Planck PR4 convergence maps on their overlap (joint sky fractions 0.05–0.08 and 0.13–0.23; ACT mask squared on the κ side as the release prescribes, Planck release mask; Gaussian covariance from the measured auto-spectra) gives b from $\langle m\kappa_{\text{CMB}} \rangle = b \int_{\text{slice}} W W_{\text{CMB}} P d\chi / \chi^2$ without any shot-noise model (Figure 6, Table 1). Every slice is detected in cross-correlation at 4–14 σ per map. Against ACT, nine of the eleven biases agree with the auto-spectrum values within 1–2 σ ; against Planck, nine. The ELG of the 0.8–1.1 slice are 35–37 per cent low in both cross-spectra (2.8 σ each), and the two ratios of BOSS 0.4–0.6 (0.74, 0.86) and of QSO 0.8–1.1 (0.77, 1.37) are inconsistent with each other. The templates use the auto-spectrum biases; the deflection validation below measures the net effect of all of them.

Wiener combination and the deflection template. Within a slice the tracer maps (in convergence units, $m/\hat{b}$) are combined with the weights $\mathbf{w}_\ell = \mathbf{C}_\ell^{-1} \mathbf{s}_\ell$ of the model covariance $\mathbf{C}_{kl} = \mathbf{S}_\ell + \mathbf{N}_{kl}$ ($\mathbf{S}_\ell$ the slice’s convergence spectrum, $\mathbf{N}$ the shot-noise matrix): inverse-shot-noise weighting of the tracers, Wiener-suppressed where noise dominates. Two features of the enlarged tracer set are handled here. The footprints differ (a BOSS slice covers a sky fraction of 0.22–0.24 against 0.17 for the DR1 LRGs, and their union 0.29–0.31), so the weights are computed per *coverage class*, the subset of tracers covering a pixel: inside each class the filtered sum is the conditional expectation of $\kappa_{\text{Ly}\alpha}$ given the tracers actually present, which is what keeps the amplitude estimator of Section 3.2 normalised to $A = 1$ without further calibration; the slice mask is the union of the tracer masks. And BOSS and DESI share objects (34 per cent of the BOSS galaxies at 0.1–0.6 are also BGS or LRG targets, 25 per cent at 0.6–0.8), so their shot noise is correlated: the off-diagonal $\mathbf{N}_{kl}$ is computed from a 1-arcsec positional match as $\sum_{\text{shared}} (u_k w_k)(u_l w_l) \Omega_{\text{pix}}^2 / \Omega_{\text{joint}}$, a noise correlation of 0.34 for BGS $\times$ BOSS and LRG $\times$ BOSS at $z < 0.6$ and 0.11 at 0.6–0.8 (the DESI tracers share fewer than 10^{-3} of their objects). The slices are independent, so the sum of the filtered slice maps is the conditional expectation of $\kappa_{\text{Ly}\alpha}$ given all tracers; its deflection, $\boldsymbol{\alpha}^T = \nabla \phi$ with $\phi_{LM} = 2\kappa_{LM}/[L(L+1)]$, evaluated at the sightlines (`healpy` derivatives at inside 1024), is the combined template. Each slice is also fitted on its own and the six slice amplitudes are combined with their joint jackknife covariance as a cross-check.

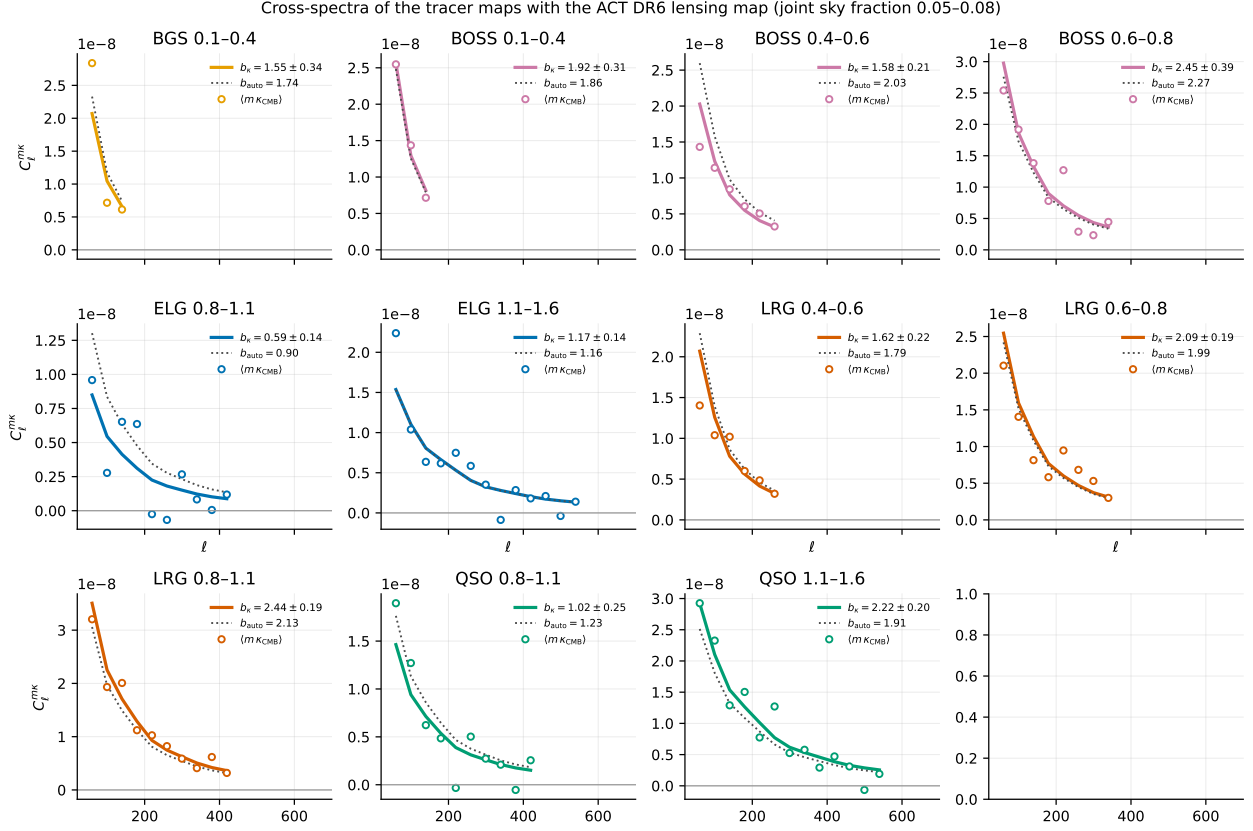


Figure 6: Cross-spectra of the unit-bias tracer maps with the ACT DR6 lensing map (points, NaMaster bandpowers with Gaussian errors) with the theory scaled by the cross-correlation bias (solid) and by the auto-spectrum bias (dotted).

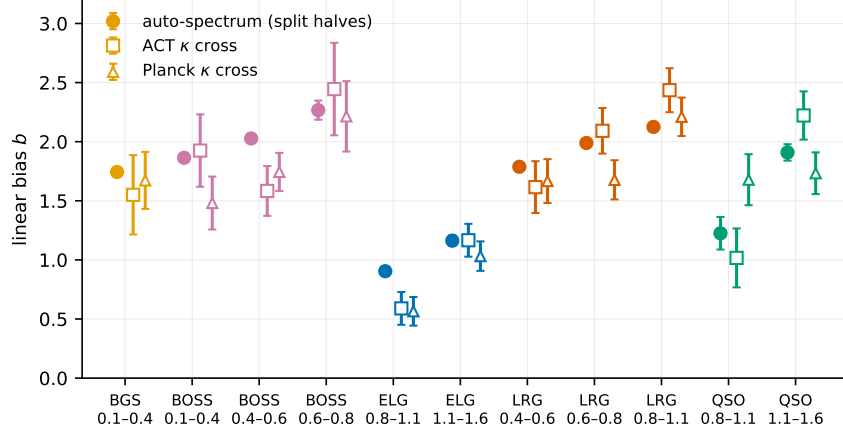


Figure 7: The three bias estimates of Table 1: auto-spectrum (filled), ACT cross (squares) and Planck cross (triangles).

template	A_L (ACT)	A_L (Planck)	A_L, κ map (ACT)	A_L, κ map (Planck)	B/E (ACT / Planck)
slice 0.1–0.4	0.98 ± 0.10	0.90 ± 0.08	1.02 ± 0.10	0.89 ± 0.08	$-0.024 / 0.046$
slice 0.4–0.6	0.82 ± 0.08	0.81 ± 0.07	0.86 ± 0.08	0.87 ± 0.06	$0.001 / 0.006$
slice 0.6–0.8	1.00 ± 0.08	0.89 ± 0.07	1.01 ± 0.08	0.87 ± 0.07	$-0.013 / -0.018$
slice 0.8–1.1	0.91 ± 0.07	0.86 ± 0.06	0.96 ± 0.07	0.94 ± 0.07	$0.043 / 0.043$
slice 1.1–1.6	0.87 ± 0.08	0.80 ± 0.07	1.04 ± 0.08	0.84 ± 0.07	$-0.062 / -0.007$
combined	0.99 ± 0.04	0.92 ± 0.03	1.01 ± 0.04	0.92 ± 0.03	$-0.009 / 0.004$
band (combined)	A_L (ACT)	A_L (Planck)	expected ratio to $\kappa_{\text{Ly}\alpha} \times \kappa_{\text{CMB}}$	measured (ACT)	measured (Planck)
40–100	0.87 ± 0.08	0.81 ± 0.06	0.56	0.49 ± 0.05	0.42 ± 0.03
100–200	0.94 ± 0.06	0.88 ± 0.05	0.42	0.40 ± 0.03	0.33 ± 0.02
200–300	1.12 ± 0.08	0.98 ± 0.08	0.30	0.34 ± 0.02	0.26 ± 0.02
300–400	1.10 ± 0.11	0.99 ± 0.10	0.23	0.25 ± 0.02	0.20 ± 0.02
400–500	1.06 ± 0.12	1.00 ± 0.13	0.18	0.19 ± 0.02	0.16 ± 0.02
science	0.99 ± 0.04	0.92 ± 0.03	0.29	0.29 ± 0.01	0.26 ± 0.01

Table 2: Deflection templates against CMB lensing. Top: the amplitude of the deflection E-mode cross-spectrum over the science window relative to the prediction for each template (expected 1), the same for the convergence template itself (spin 0), and the B/E leakage. Bottom: the combined template per science band, with the ratio of the prediction to the full $\kappa_{\text{Ly}\alpha} \times \kappa_{\text{CMB}}$ spectrum and the measured ratio to it.

Validation of the deflection templates against CMB lensing. The product the estimator consumes is the band-filtered deflection of the Wiener-filtered template. That field is built on the sphere for every science band and for the whole science window ($40 \leq L \leq 500$), and cross-correlated as a spin-1 field with the ACT and Planck convergence maps with NaMaster (spin-1 $\times$ spin-0 bandpowers of width 40, decoupled from the tracer mask and the CMB mask; the ACT map is treated as already carrying its mask squared, as the release prescribes; errors from the Gaussian covariance with the measured auto- and cross-spectra): the E-mode of the deflection carries the signal and its B-mode is mask leakage of the E-mode, which is quoted as the ratio B/E. The prediction is not $C_L^{\kappa_{\text{Ly}\alpha} \kappa_{\text{CMB}}}$: the template estimates the part of $\kappa_{\text{Ly}\alpha}$ traced by objects at $z < 1.75$, Wiener-suppressed where they are noisy, so for $T = \sum_s \sum_c \mathcal{W}_c^s \sum_{k \in c} W_k^c * m_k$ the expectation of a pseudo-spectrum against κ_{CMB} on the sky weight M (the product of the two masks) is $\langle T \kappa_{\text{CMB}} \rangle_L = \sum_s \sum_c \frac{\langle \mathcal{W}_c^s M \rangle}{\langle M \rangle} \sum_{k \in c} W_k^c(L) C_L^{(\ell_s, c)} p_L$, with $C^{(\ell_s, c)}$ the Limber cross-spectrum of the slice’s contribution to $\kappa_{\text{Ly}\alpha}$ with κ_{CMB} and p_L the pixel window; the class fractions are taken inside the overlap with each CMB map and normalised by the overlap area, because NaMaster attributes the cross-spectrum to the whole field mask while a slice’s tracers may cover only part of it (the slices above $z = 0.8$ are DESI-only, the combined footprint includes BOSS-only sky). The deflection E-mode then reads $C_L^{E\kappa} = 2/\sqrt{L(L+1)} F(L) \langle T \kappa_{\text{CMB}} \rangle_L$ for a band window F . The amplitude A_L of the measured cross relative to this prediction is expected to be one when the template chain (biases, shot noise, Wiener weights) is right, whatever the source redshifts; the ratio to the full $\kappa_{\text{Ly}\alpha} \times \kappa_{\text{CMB}}$ spectrum, which is what a naive comparison would use, is 0.16–0.57 depending on the band and is tabulated for reference. Table 2 and Figure 8 give the result (`report/stageb/deflection_cmb_check.json`, `cmb_map_checks.json`).

Every template is detected at 5–30 σ per map, the B-modes are at most 6 per cent of the E-modes, and the per-band χ^2 are acceptable. For the combined template the deflection amplitude is 0.99 ± 0.04 against ACT and 0.92 ± 0.03 against Planck, the convergence template gives 1.01 ± 0.04 and 0.93 ± 0.03 , the five slices lie between 0.80 and 1.00, and the per-band amplitudes rise mildly with L (0.88/0.82 at 40–100 to 1.06/1.00 at 400–500). The two CMB maps agree: on their common sky (a fraction 0.093) the convergence template gives 1.01 ± 0.04 (ACT) and 0.93 ± 0.06 (Planck), and by region the Planck amplitude is 0.90–0.95 everywhere (north and south of the DESI imaging boundary, both Galactic caps), the ACT amplitude 0.92 in the SGC and 1.11 ± 0.06 in the NGC. Two mistakes were found and removed on the way to these numbers, and are recorded because they are the kind a later reader would want to know about: multiplying the ACT map by its mask squared a second time (3 per cent low), and normalising the class fractions of the prediction by each slice’s own footprint rather than by the overlap, which over-predicted the cross wherever the high-redshift,

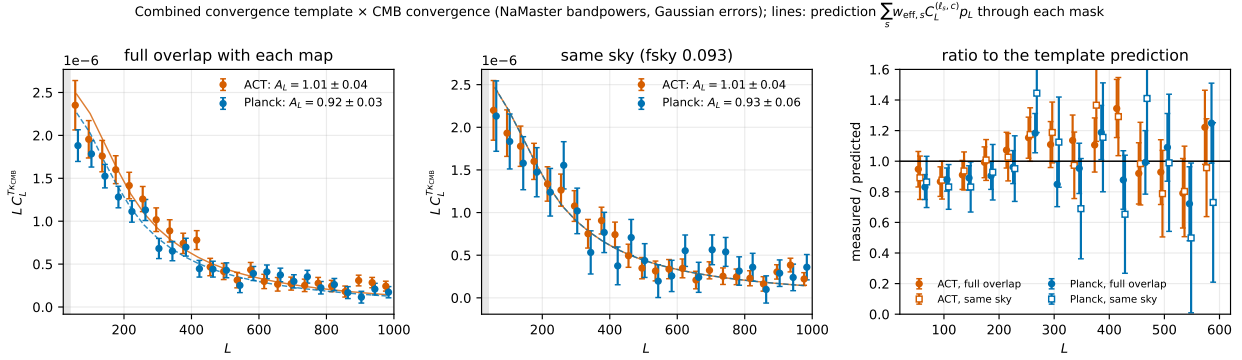


Figure 8: The combined convergence template against the ACT (orange) and Planck (blue) convergence maps: NaMaster bandpowers with Gaussian errors on the full overlap with each map (left) and on their common sky (middle), with the prediction described in the text pushed through each mask pair (lines), and the ratio of measurement to prediction (right).

DESI-only slices are absent (the northern sky) and produced an apparent 0.71 against Planck with a strong regional dependence. With the correct prediction the previous DESI-only template also gives 0.94 ± 0.07 against Planck with no regional dependence. A direct ACT $\times$ Planck convergence cross-spectrum is not a usable calibration test: the two reconstructions share the same CMB modes and their noise is correlated, so the cross exceeds $C_L^{\kappa\kappa}$ by a factor of two with a χ^2 of 2300; the ACT auto-spectrum, however, matches $C_L^{\kappa\kappa} + N_L$ to 4 per cent. The remaining 5–8 per cent deficit against Planck and the mild rise with L are at the level expected from a clustering amplitude below the fiducial model’s (the cross-to-auto bias ratio scales as $\sigma_8^{\text{true}}/\sigma_8^{\text{fid}}$) and from residual large-scale systematic power in the tracer auto-spectra; they are within the 20 per cent systematics budget appropriate at $\sigma(A) \simeq 0.5$ and no correction is applied.

3.4 Why not a CMB-lensing template

The same estimator cross-correlated with κ_{CMB} measures lensing plus the intrinsic forest–forest–density three-point function of the forest’s own redshift range. Its linear part, the response of the small-scale forest power to long-wavelength modes, can be removed with a kernel-matched template built from the $z > 2$ quasars (the “deprojection” of the main report), but the second-order part, $\langle m_p m_q T \rangle$ with m the long-wavelength component of the forest and T a non-Gaussian tracer template, cannot be removed by one coefficient: on 400 mock realisations it biased the deprojected amplitude by -0.41 ± 0.18 with no lensing and no response present. The lensing signal, the response and this term all act through the same long modes at the pair separations used, so the term has to be modelled, not filtered. A tracer at $z < 1.8$ shares no density modes with the forest at $z > 2.1$: its correlation with the pair products is lensing only, plus the magnification of the sightline quasars (absorbed by the mean field) and shared sky systematics.

4 Validation on mocks

The mocks are flat-sky FFT realisations of a $20^\circ \times 20^\circ$ patch: a Gaussian matter field on a $2 h^{-1}\text{Mpc}$ grid over the forest box $2.1 < z < 3.0$ (plus margins) from which the forest is a linear, redshift-space-distorted filter with $b_F = -0.13$, $\beta_F = 1.6$ and a small-scale cut-off; 22 sightlines per deg^2 with pixel noise $P_N = 0.33 h^{-1}\text{Mpc}$ (log-normal scatter across forests) and per-forest continuum projection; the ACT DR6 mask cut-out and a smooth completeness pattern; lognormal quasars behind the slab draw the sightlines. The foreground of $\kappa_{\text{Ly}\alpha}$ (and of κ_{CMB}) is generated in the five tracer slices as Gaussian projected-density maps with the exact Limber covariance between the density and the two convergence contributions, plus the uncovered rest; each slice hosts 2-D lognormal Poisson tracers of the projected density smoothed transversely by $3 h^{-1}\text{Mpc}$ (a lognormal of the raw $0.7 h^{-1}\text{Mpc}$

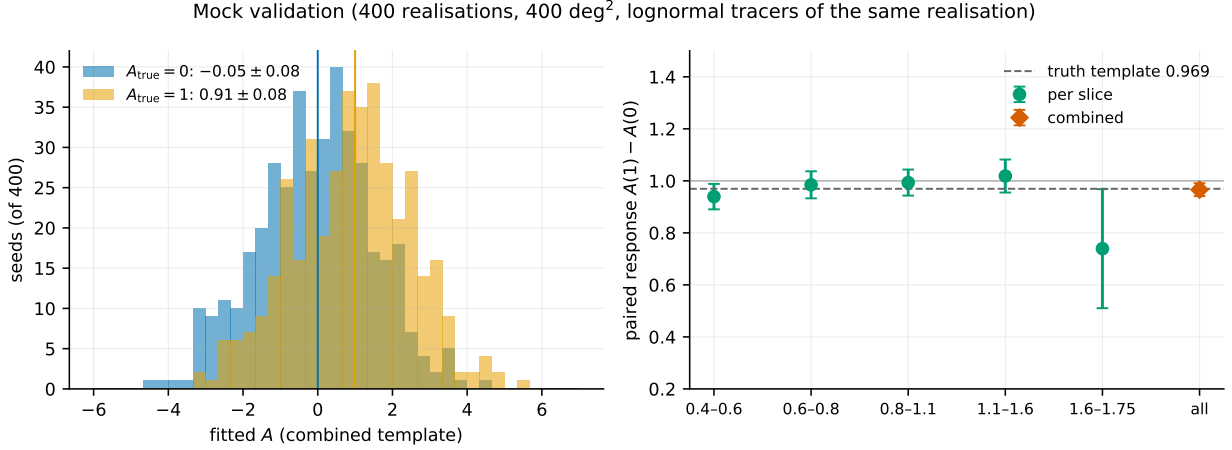


Figure 9: Mock validation on 400 realisations. Left: the combined-template amplitude for an unlensed ($A_{\text{true}} = 0$) and a lensed ($A_{\text{true}} = 1$) forest of the same realisations. Right: the paired response $A(1) - A(0)$ per slice and combined, against the truth template.

cells is absurdly non-Gaussian), with DR1-like biases and densities. The forest is lensed by the full $\kappa_{\text{Ly}\alpha}$ (sampling at the displaced positions), and every realisation carries both a lensed and an unlensed forest.

The protocol (GATES.md v7) was frozen before the scale-1 campaign: 400 fresh realisations, templates built from each realisation’s own tracers exactly as on the data (bias from the auto-spectrum, Wiener combination, sum over slices), and the gate “an unbiased detection against a lognormal redshift tracer”. Figure 9 and Table 3 give the result: the combined null and recovery are unbiased, the paired response (the same realisation lensed and unlensed, free of sample variance) is 0.966 ± 0.025 , equal to the response of the truth template (0.970 ± 0.018 ; the 3 per cent deficit of the noisy sparse sample was already seen with the CMB template and is a property of the noisy forest, not of the template chain), the per-slice responses are unity within errors, a template from another realisation gives zero, the curl is zero, the jackknife error matches the ensemble scatter, and the noise-free injection expectation (the bookkeeping test of signs, cuts and band basis) has odd slope 0.98. The only rows of the protocol that fail are the tracer-bias precision rows: the auto-spectrum bias of a lognormal tracer differs from the generator’s by 1–3 per cent, resolved at $N = 400$ against a 2-SEM rule that amounts to 0.4 per cent; the effect on A is the same 1–3 per cent per slice.

What the corrected response table costs. The acceptance campaign above was run with the two-parameter table of iteration 5. The correction of Equation (3) is fitted to the same pixel pairs the estimator weights, so it can only lose response, never gain it, on a mock whose model shape is already correct. Forty of the 400 realisations were refitted with each variant of the correction and with the two-parameter table, everything else identical (`code/pipeline/validate_xi_correction.py`, `report/xi_correction_mocks_variants.json`); the paired responses are

response table	fitted parameters	paired response	ratio	DR1 amplitude gain
two-parameter (iteration 5)	2	0.974 ± 0.071	—	1
bicubic correction (adopted)	22	0.988 ± 0.073	1.014 ± 0.012	1.077
medium knots	36	0.935 ± 0.068	0.960 ± 0.018	1.099
fine knots	48	0.873 ± 0.064	0.896 ± 0.023	1.100
fine knots, no same-wavelength term	42	0.900 ± 0.066	0.924 ± 0.022	1.067

The ratios are paired differences on the same realisations, so their errors are much smaller than those of the responses themselves. They follow the attenuation $\langle W \text{Var}(\delta g) \rangle / \langle W g^2 \rangle$ computed from

statistic (400 realisations, 400 deg ²)	value	requirement
combined null, $A_{\text{true}} = 0$	-0.05 ± 0.08	$\leq 2 \text{ SEM}; 95\% \text{ bound } 0.21 (\leq 0.5)$
combined recovery, $A_{\text{true}} = 1$	0.91 ± 0.08	$\leq 2 \text{ SEM}; 95\% \text{ bound } 0.24 (\leq 0.5)$
paired response $A(1) - A(0)$	0.966 ± 0.025	$ \cdot - 1 \leq 2 \text{ SEM}$
truth-template paired response	0.970 ± 0.018	report
per-slice responses (0.4–0.6 ... 1.6–1.75)	0.94, 0.99, 0.99, 1.02 (± 0.05); 0.74 ± 0.23	each within 2 SEM
other-realisation template	0.01 ± 0.08	$ \cdot \leq 2 \text{ SEM}$
curl (combined)	-0.07 ± 0.10	$ \cdot \leq 2 \text{ SEM}$
scatter / RMS jackknife	0.96	0.7–1.3
normalisation slope on $A \in \{0, \frac{1}{2}, 1, 2\}$	0.97 ± 0.06	within 2 SEM of 1
response on – off	0.08 ± 0.09	$ \cdot \leq 2 \text{ SEM}$
injection expectation, odd slope at $ A = 0.25$	0.980	1 ± 0.05
tracer biases $b_{\text{fit}}/b_{\text{true}}$	$0.97\text{--}1.01 (\pm 0.002\text{--}0.004)$	2 SEM (fails for 6 of 8)
per-seed scatter of the combined amplitude	1.56	(CMB deprojected estimator: 3.7)

Table 3: Iteration-7 acceptance (`report/lowz_validation.md`).

the least-squares covariance of the fit, which gives 0.7, 2.4 and 4.1 per cent for the three knot sets on the same mock: the right ordering, and about 2.5 times smaller than measured, the difference being the correlation between neighbouring cells that the diagonal pair weights ignore. The adopted correction costs nothing measurable, and the same calculation on the DR1 counts gives 0.25 per cent before that factor. The mock forests carry no sky residual, so the same-wavelength term has nothing real to fit there; that it is not the source of the attenuation is visible in the last row, which removes it and recovers only a third of the loss.

5 Measurement on DR1

The full run (workstation, 78 minutes, 34 GB) fits the evolving forest correlation, builds the pair catalogue (6.5×10^6 sightline pairs), evaluates the five slice templates and the combined template at the 427 888 sightlines, and fits the eleven-component basis. Table 4 and Figure 10 give the result:

$$A = 0.35 \pm 0.47 \text{ (combined template, jackknife; Fisher scale 0.38)}, \quad A < 1.13 \text{ at 95 per cent (one-sided)}. \quad (6)$$

The five slices are individually consistent with zero and their jackknife-covariance combination, 0.21 ± 0.46 , agrees with the combined template. The nulls: the curl amplitude is -0.58 ± 1.60 ; forty Gaussian random realisations of the combined map’s own spectrum through the tracer mask, fitted exactly like the data, give a mean of -0.06 ± 0.07 and a scatter of 0.45 against the RMS jackknife error 0.48 (the jackknife is 7 per cent conservative); the noise-free injection expectation reads 1.026. The five band amplitudes are 0.28 ± 0.90 , 0.28 ± 0.72 , -0.47 ± 0.99 , 0.28 ± 1.33 and 3.15 ± 1.56 for $L = 40\text{--}100$ to $400\text{--}500$: as in the previous versions, the top band is the one that pulls the common amplitude up.

Where the improvement comes from. The measurement has changed in six ways since the first version of this note: the refined response table (Section 3.2), the bands extended from $L = 300$ to 500, region B, the redshift-evolving correlation table together with the enlarged tracer set (BGS and BOSS, NaMaster biases, per-class Wiener weights), and finally the whole DR1 grid below $z = 3$ with the source plane at the weighted mean pixel redshift. Running each step on the same data separates them:

template	A	jackknife	Fisher
combined (five slices)	0.346	0.474	0.379
slice 0.1–0.4	0.83	1.03	0.82
slice 0.4–0.6	−0.51	0.73	0.73
slice 0.6–0.8	0.11	1.04	0.80
slice 0.8–1.1	1.39	0.95	0.83
slice 1.1–1.6	0.02	1.43	1.29
jackknife combination of the slices	0.208	0.461	
random templates (40)	$−0.06 \pm 0.07$	scatter 0.45	RMS jackknife 0.48
combined, previous version ($2.1 < z < 3$, six slices)	0.271	0.519	0.398
combined, first version ($2.1 < z < 3$, five slices, non-evolving table)	0.182	0.663	0.514
combined, region A only, $L < 300$	−0.101	0.754	0.646

Table 4: DR1 lensing amplitude, single slab $1.96 < z < 3.0$ (every DR1 pixel below $z = 3$), with both delta regions, bands to $L = 500$, the redshift-evolving correlation table, the source plane at $z_{\text{eff}} = 2.35$ and the eleven-tracer templates to $z = 1.6$ (`report/stageb/dr1_lowz_v7.md`); the last rows are the previous versions for comparison.

forest	bands, table, tracers	A	jackknife	Fisher	jk/Fisher	sightline
A, $2.1 < z < 3$	$L < 300$	−0.101	0.754	0.646	1.17	3.0
A, $2.1 < z < 3$	$L < 500$	0.170	0.735	0.573	1.28	3.0
A + B, $2.1 < z < 3$	$L < 500$	0.182	0.663	0.514	1.29	5.0
A + B, $2.1 < z < 3$	$L < 500$, evolving table, 12 tracers	0.271	0.519	0.398	1.30	5.0
A + B, $2.1 < z < 3$	$L < 500$, flat table, 12 tracers	0.274	0.537	0.417	1.29	5.0
A + B, $1.96 < z < 3$	$L < 500$, evolving table, 11 tracers, z_{eff}	0.346	0.474	0.379	1.25	6.5

The extra bands cut the Fisher error by 11 per cent and the jackknife error by 3, and move the central value by $+0.27$ (0.37σ). Region B cuts both errors by a further 10 per cent at an unchanged central value. The enlarged tracer set cuts the Fisher error from 0.514 to 0.417 and the jackknife error from 0.663 to 0.537, and moves the central value by $+0.09$; the evolving table on top of it changes the central value by $−0.003$ and trims the error by 3 per cent. Extending the forest from $z = 2.1$ down to the blue end of the grid at $z = 1.96$ adds 29 per cent more pixel pairs and cuts the errors by a further 9 per cent (jackknife) and 5 per cent (Fisher), at a central value $+0.08$ higher. The ratio of the jackknife error to the Fisher scale, 1.25, is stable: the high- L bands scatter more than the Gaussian estimate expects, and the random-template scatter, 0.45 against the RMS jackknife 0.48, again says the jackknife is the conservative one.

With the two-parameter table of Section 3.2 the A-only, $L < 300$ run gives $A = 0.01 \pm 0.67$. The noise-free injection expectation reads 1.042 rather than 1.014, and that is a property of the test rather than of the estimator: the injection displaces the whole table, including the same-wavelength term, which the kernel deliberately omits because it is not lensed. Repeating the injection with that term removed from the injected correlation returns 1.014 (`report/stageb/injection_same_wavelength_v3.json`), so the bookkeeping of signs, cuts and the band basis is unchanged.

6 The quasar–forest cross-correlation

The lensing information in the forest is not confined to the forest auto-correlation. For a quasar at (θ_q, χ_q) and a forest pixel on another sightline at (θ_p, χ_p) , lensing remaps both positions, so the observed cross-correlation is ξ_{qF} at the true separation,

$$\langle \delta_F(p) | \text{quasar at } q \rangle = \xi_{qF}(r_\perp, r_\parallel) + \frac{\partial \xi_{qF}}{\partial r_\perp} \chi \hat{\theta}_{qp} \cdot (\alpha_q - \alpha_p), \quad (7)$$

the pair form of Equation (1) with $(a, b) \rightarrow (q, p)$; the estimator of Section 3.2 applies with the quasar as the first member of every pair, $w_q = 1$, and the product $\delta_F(p)\delta_F(q)$ replaced by $\delta_F(p)$.

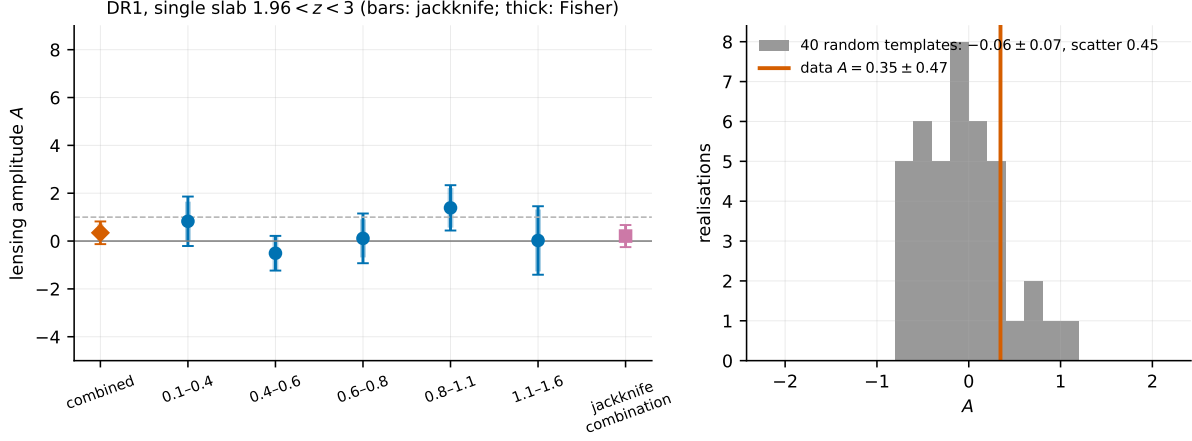


Figure 10: Left: the DR1 amplitudes per slice and combined (thin bars: jackknife, thick: Fisher) and the jackknife-covariance combination of the slices; the dashed line is the Λ CDM value. Right: the forty random-template amplitudes with the data value.

Magnification bias modulates the observed quasar density by $(5s - 2)\kappa$, but the mean field is subtracted per *observed* quasar, so that modulation contributes nothing at linear order in a foreground template (its residual, $(5s - 2)^2 \langle \kappa_{fg} \kappa^T \rangle \text{Cov}(\kappa_{fz}, \delta_F)$, is second order) and is dropped. The quasars are the DR1 quasar catalogue at $1.9 < z < 3.1$, one entry per target (583 148 quasars, 377 715 of them the quasars of forests in the sample); a quasar’s own sightline is excluded from its pairs; the templates are evaluated at the quasar positions as at the sightlines, and the jackknife, curl and random templates are as before. The radial separation is signed, $r_{\parallel} = \chi_p - \chi_q$.

The cross-correlation model. ξ_{qF} is measured in $1 h^{-1}\text{Mpc}$ cells over $r_{\perp} < 40$, $|r_{\parallel}| < 40 h^{-1}\text{Mpc}$ and seven bins of the pair mean redshift (3.6×10^7 quasar–sightline pairs), and fitted with an independent model built like Equation (4):

$$\xi_{qF}(r_{\perp}, r_{\parallel}, z) = b_q(z) b_F(z) \left[\frac{D(z)}{D(z_{\text{ref}})} \right]^2 [T_0 + (\beta_q + \beta_F)T_1 + \beta_q \beta_F T_2] (r_{\perp}, r_{\parallel} - \Delta r_{\parallel}) \otimes \mathcal{N}(\sigma_{\parallel}) \\ + x^{\gamma_S} \sigma(r) S(r_{\perp}, r_{\parallel}), \quad b_q(z) = b_q x^{\gamma_q}, \quad \beta_q(z) = \frac{f(z)}{b_q(z)}, \quad (8)$$

where the basis tables are the same three Hankel transforms pushed through the DESI pixel windows and the *one-sided* continuum projection (the projector acts on the forest pixels only, averaged over 400 real forests with the quasar distance stepped across and beyond each forest), the forest side ($b_F = -\sqrt{b_F^2}$, β_F and their power laws) is held at the auto-correlation base fit, the quasar bias and its evolution are free with β_q set by the growth rate of the cosmology, $\Delta r_{\parallel}$ is the quasar redshift offset (the observed correlation is the true one displaced along the line of sight) and $\sigma_{\parallel}$ a Gaussian redshift-error smoothing, both applied to the fine tables before the cell averaging, and the spline correction allows odd terms in $r_{\parallel}$. The base model fits with $b_q = 3.20$ at $z_{\text{ref}} = 2.4$ evolving as $(1 + z)^{1.7}$ (2.6 at $z = 2$, 4.2 at $z = 3$; the DESI quasar-clustering relation gives 3.8 at 2.4, 15 per cent higher, the difference being absorbed by the fixed b_F), $\beta_q(z_{\text{ref}}) = 0.30$, $\Delta r_{\parallel} = 0.44 h^{-1}\text{Mpc}$ and $\sigma_{\parallel} = 4.9 h^{-1}\text{Mpc}$, with $\chi^2 = 6915$ over 11 340 cells; with the base parameters held there, the 20-coefficient correction takes χ^2 to 5729, uniformly across the redshift bins (0.46–0.65 per cell), and the measured-to-model amplitude per redshift bin is within 4 per cent of one. Figures 11 and 12 are the QA plots: the model follows the line-of-sight profiles at every $r_{\perp}$, offset and smoothing included, and the residual map in the fit range is featureless.

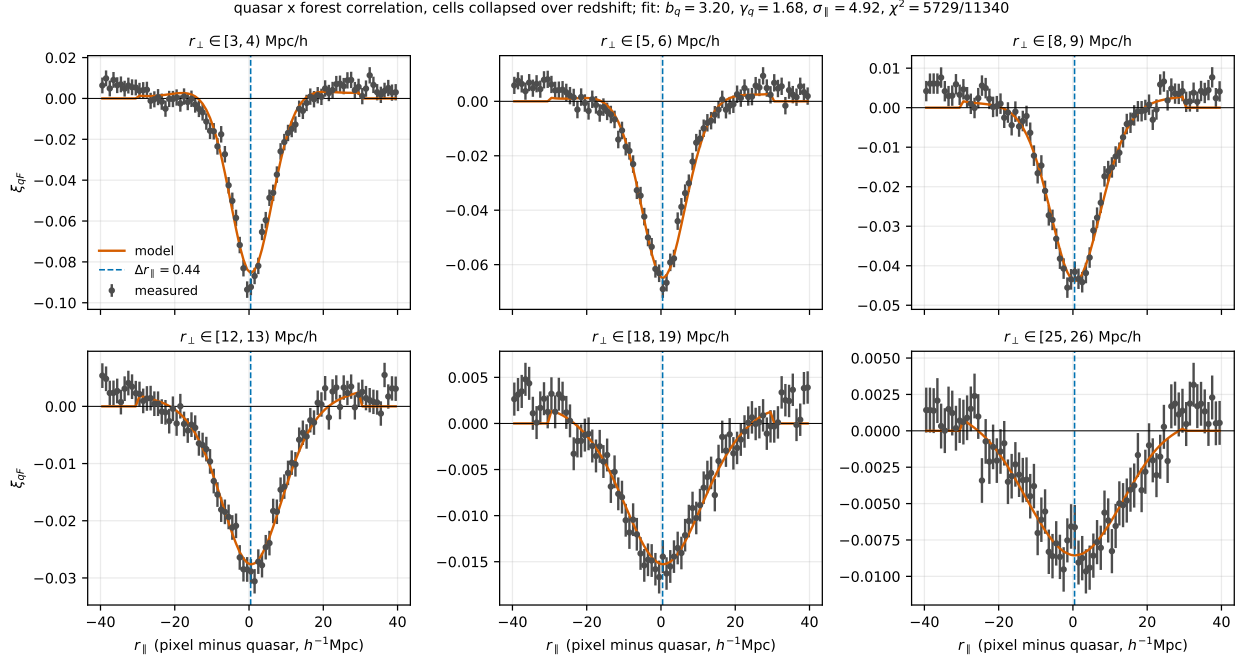


Figure 11: The quasar–forest correlation in line-of-sight profiles at six transverse separations, collapsed over redshift (points, with the pair-weight errors), and the fitted model of Equation (8) (line); the dashed line marks the fitted quasar redshift offset.

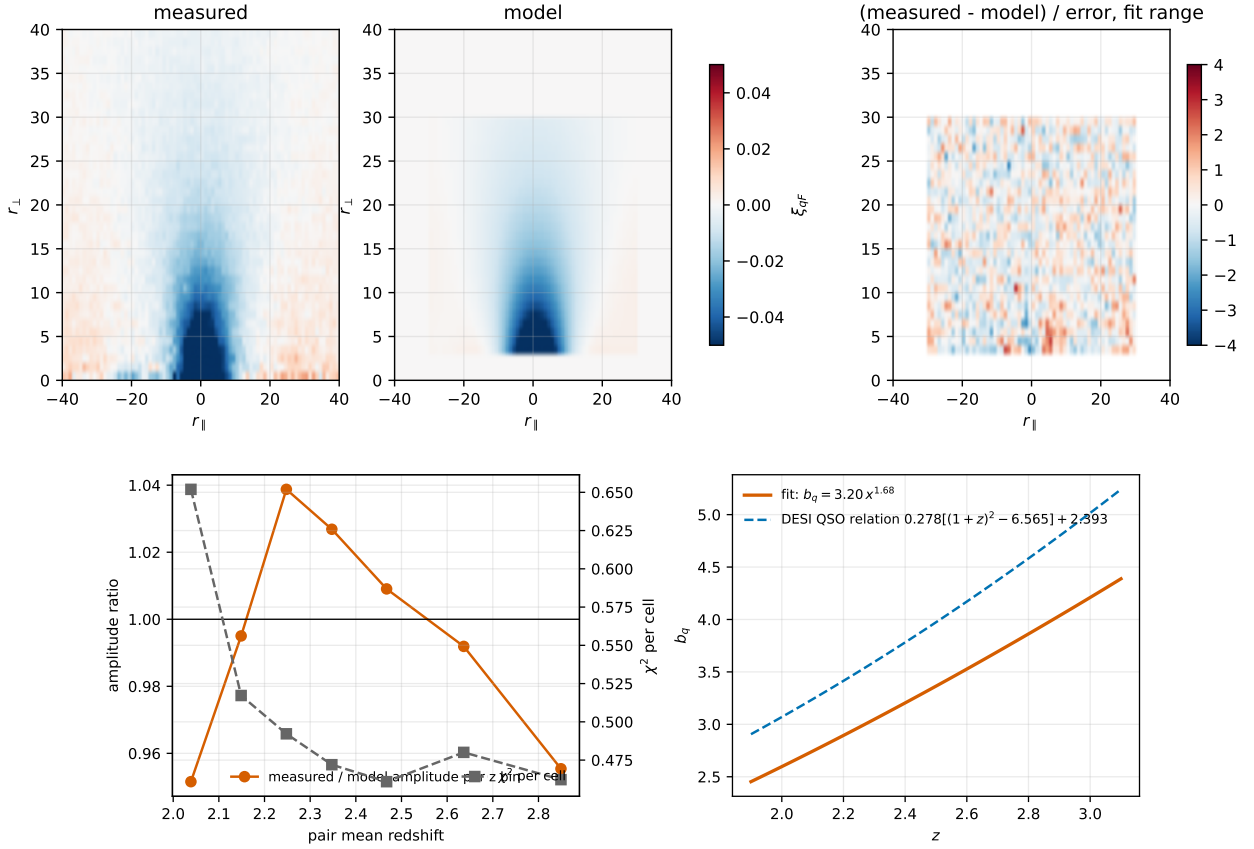


Figure 12: Top: the measured and modelled cells and their residual in units of the error over the fit range. Bottom: the measured-to-model amplitude and the χ^2 per cell in each redshift bin, and the fitted $b_q(z)$ against the DESI quasar-clustering relation.

	A (forest $\times$ forest)	A (quasar $\times$ forest)	corr.	combined	cross weight (%)
combined template, $40 \leq L \leq 500$	0.35 ± 0.47	0.61 ± 0.52	-0.01	0.47 ± 0.35	46
band 40–100	0.28 ± 0.90	-0.34 ± 1.37	+0.14	0.12 ± 0.80	27
band 100–200	0.28 ± 0.72	0.01 ± 0.93	+0.01	0.18 ± 0.58	37
band 200–300	-0.47 ± 0.99	1.16 ± 1.33	+0.07	0.09 ± 0.82	34
band 300–400	0.28 ± 1.33	1.41 ± 1.62	+0.12	0.72 ± 1.09	39
band 400–500	3.15 ± 1.56	3.82 ± 2.05	+0.03	3.39 ± 1.26	36
pairs at 1.96–2.25	0.26 ± 0.67	0.32 ± 1.02	+0.14	0.27 ± 0.60	27
pairs at 2.25–2.55	0.59 ± 0.67	1.43 ± 0.94	+0.16	0.85 ± 0.59	30
pairs at 2.55–3	0.07 ± 0.88	-0.31 ± 1.14	+0.26	-0.05 ± 0.78	33
slice 0.1–0.4	0.83 ± 1.03	0.42 ± 1.28	+0.22	0.68 ± 0.89	36
slice 0.4–0.6	-0.51 ± 0.73	0.63 ± 1.07	+0.23	-0.21 ± 0.66	27
slice 0.6–0.8	0.11 ± 1.04	1.25 ± 1.22	+0.14	0.58 ± 0.85	41
slice 0.8–1.1	1.39 ± 0.95	0.57 ± 1.14	+0.32	1.09 ± 0.84	36
slice 1.1–1.6	0.02 ± 1.43	-0.59 ± 1.92	+0.28	-0.16 ± 1.29	31

Table 5: Forest auto- and quasar–forest cross-correlation lensing amplitudes with their jackknife errors, the jackknife correlation of the two, their optimal combination and the weight of the cross-correlation in it (`report/stageb/auto_cross_combination.json`): overall, per template band, per range of the pair mean redshift, and per tracer slice.

Result and combination. On 9.5×10^6 quasar–sightline pairs within $30 h^{-1}\text{Mpc}$ the cross-correlation gives

$$A_{qF} = 0.61 \pm 0.52 \text{ (jackknife; Fisher scale 0.45)}, \quad (9)$$

with curl -0.71 ± 2.18 , forty random templates at -0.10 ± 0.10 with scatter 0.66 against the RMS jackknife error 0.62, the five slices consistent and their jackknife combination 0.58 ± 0.52 . Its precision is close to that of the forest auto-correlation (0.35 ± 0.47), and the two are essentially independent: the jackknife correlation of the two amplitudes is -0.01 (they share the forest pixels but not the noise that dominates either). Table 5 combines them with the joint jackknife covariance, overall, per template band, per redshift range of the pairs and per tracer slice; the cross-correlation carries 46 per cent of the weight overall and 27–39 per cent per band. The combination is

$$A = 0.47 \pm 0.35, \quad A < 1.04 \text{ at 95 per cent (one-sided)}. \quad (10)$$

The redshift split shows the same picture in both statistics: the middle range $2.25 < z < 2.55$ is highest (0.85 ± 0.59 combined), the ends consistent with zero. The kernel dependence of the cross-correlation amplitude is bounded below in Section 7.

6.1 An alternative band set to $L = 1000$

The template validation of Section 3.3 holds to $L = 600$ and beyond, so the analysis was repeated with the science bands doubled in width and extended to the template’s band limit: 40–200, 200–400, 400–600, 600–800 and 800–1000, with the deflection evaluated at $n_{\text{side}} = 2048$. The deflection validation with these bands gives 0.97 ± 0.04 (ACT) and 0.93 ± 0.03 (Planck) over $40 \leq L \leq 1000$, and per band the values in the last column of Table 6: the templates hold to $L = 1000$, the top two bands being high by $2\text{--}3\sigma$ against one map each rather than low. Table 6 gives the amplitudes (`report/stageb/dr1_lowz_v7b`, `dr1_qso_v1b`, `auto_cross_combination_wide.json`). The wider range cuts the error of the auto-correlation by 15 per cent at an unchanged central value (0.34 ± 0.40), that of the cross-correlation by 13 per cent with a $+0.26$ shift (0.87 ± 0.45), and the combination becomes

$$A = 0.58 \pm 0.31 \quad (40 \leq L \leq 1000), \quad \text{against } 0.47 \pm 0.35 \text{ for } 40 \leq L \leq 500. \quad (11)$$

Both statistics put their excess in the 400–800 range (combined 2.0 ± 0.9 and 1.6 ± 1.3), the same feature as the 400–500 band of the narrow set, and the 800–1000 band is a null (-0.8 ± 1.6); the redshift split is flat within its errors. The nulls of the wide set: curl 0.03 ± 1.89 (auto) and -0.43 ± 2.64

	A (forest×forest)	A (quasar×forest)	combined	template vs CMB (ACT / Planck)
narrow set, $40 \leq L \leq 500$	0.35 ± 0.47	0.61 ± 0.52	0.47 ± 0.35	
wide set, $40 \leq L \leq 1000$	0.34 ± 0.40	0.87 ± 0.45	0.58 ± 0.31	
band 40–200	0.20 ± 0.55	0.07 ± 0.69	0.15 ± 0.44	0.90 ± 0.05 / 0.85 ± 0.04
band 200–400	0.02 ± 0.81	1.54 ± 0.95	0.66 ± 0.63	1.10 ± 0.06 / 0.99 ± 0.06
band 400–600	2.06 ± 1.09	1.89 ± 1.48	2.00 ± 0.90	1.03 ± 0.09 / 0.97 ± 0.10
band 600–800	1.34 ± 1.51	2.03 ± 1.86	1.60 ± 1.27	1.01 ± 0.13 / 1.40 ± 0.15
band 800–1000	-2.65 ± 1.85	4.25 ± 2.92	-0.79 ± 1.63	1.47 ± 0.17 / 1.23 ± 0.20
wide set, pairs at 1.96–2.25	0.46 ± 0.62	0.91 ± 0.86	0.60 ± 0.53	
wide set, pairs at 2.25–2.55	0.10 ± 0.56	1.42 ± 0.84	0.48 ± 0.50	
wide set, pairs at 2.55–3	0.58 ± 0.74	-0.05 ± 0.95	0.37 ± 0.65	

Table 6: The alternative band set: the auto- and cross-correlation amplitudes and their optimal combination for the narrow and the wide science windows, per wide band (with the deflection-template validation against ACT and Planck in that band) and per range of the pair mean redshift.

(cross); twenty random templates at -0.19 ± 0.08 (auto, scatter 0.37 against the jackknife 0.42) and 0.17 ± 0.14 (cross, scatter 0.64 against the jackknife 0.54, the jackknife under-estimating by 15 per cent here); injection expectation 1.024. The narrow set remains the headline because its bands are the ones the mocks validated; the wide set is the more precise measurement.

7 Caveats and systematics

At an error of 0.47 on A , systematic effects of 20 per cent are still below the noise; the following are recorded for the full-DESI measurement, the template normalisation first.

- *Slice-to-slice cross-talk.* Each slice is fitted on its own, with an 11×11 response matrix and the other slices absent from the model; strictly the fit should be joint, one $(11 \times 5)^2$ matrix, because the slice templates are not orthogonal on the pair basis. The leakage was measured for the five-slice version in `report/stageb/slice_crosstalk.json`: the response correlation between the common-science directions of two slices never exceeds 0.037, and the leakage matrix $L_{ss'} = \partial \hat{A}_s / \partial A_{s'}$ has row sums 0.97, 0.99, 0.95, 0.89, 1.06, so a separate per-slice fit returns between 0.89 and 1.06 when every slice truly has $A = 1$. Correcting the slice amplitudes by L^{-1} moves their jackknife combination from 0.353 ± 0.625 to 0.377 ± 0.650 , a 0.04σ shift. The combined-template result, which is the number we quote, is a single template in a single joint band fit and carries no slice decomposition, so it is unaffected. A joint fit is a memory-layout change rather than a hard one: `amplitude` currently holds an $(N_t, N_t, N_{\text{pair}})$ array, 124 GB at $N_t = 55$, where accumulating per jackknife region instead would need 7 MB.
- *One response table for two pair types.* With the evolving table the two pair types differ by 10 per cent in amplitude at fixed redshift (Section 3.2); by the same weight argument as before ($A \times B$ carries 22 per cent of the correlation weight and 21 per cent of the Fisher weight) the bias on A is at the 0.1 per cent level, with the shape difference not separately cancelled.
- *High- L bands.* The ratio of the jackknife error to the Fisher scale grows from 1.17 to 1.29 when the bands are extended to $L = 500$, so the two new bands scatter more than the Gaussian estimate predicts. They also carry the $+0.27$ shift in the central value. The tracer bias is measured only over $40 \leq \ell \leq 0.2\chi$, which for the nearest slice ends at $\ell \simeq 260$, so the 300–500 bands rely on extrapolating the bias; the ELG auto-spectrum already shows scale-dependent bias above $\ell \simeq 300$.
- *Tracer biases and the template normalisation.* The auto-spectrum and CMB-cross biases agree to 10–15 per cent for most tracers, and the deflection validation gives $A_L = 0.99 \pm 0.04$ (ACT) and 0.92 ± 0.03 (Planck) for the combined template, 0.80–1.00 per slice, with a mild rise with L . A template normalisation error f scales the lensing amplitude by $1/f$, so the 8 per cent

Planck deficit, if real, is a +8 per cent question for the result, within the budget; the ELG of the 0.8–1.1 slice (cross-to-auto ratio 0.65 against both maps) remain the clearest individual outlier.

- *Forest correlation model.* The correction of Equation (3) is empirical: it is fitted to the same pixel pairs that the estimator multiplies, so the kernel can follow noise in the measured correlation, which attenuates the amplitude. With the adopted 22 parameters the mock test bounds that at 1.4 ± 1.2 per cent, and the model dependence of the kernel across the projection variants is 2.0–5.3 per cent rms, or 2 per cent on A . Both will need re-examining for the full sample: the attenuation is a property of the fit, not of the sample size, and the natural fix is to estimate the covariance of the measured cells (a jackknife of the correlation measurement) and to subtract $\langle W \text{Var}(\delta g) \rangle$ from the response matrix.
- *Continuum projection.* The projection is now averaged over real forest pairs, but it remains a model of picca’s continuum fit as a per-forest weighted mean-and-slope removal, and it ignores the stacked-delta correction that picca applies across forests at fixed observed wavelength. With the corrected projection the model predicts an $r_{\perp}$ tilt over $3 < r_{\perp} < 30 h^{-1} \text{Mpc}$ that the data do not show; the empirical correction absorbs the difference, and the kernel is the same to 5 per cent whichever projection is used as the base, but the disagreement is not understood.
- *Mask coupling.* The tracer spectra and the CMB cross-spectra are NaMaster bandpowers with analytic mode coupling; the residual approximation is the Gaussian covariance and, for the deflection field, the fact that the deflection inside a fragmented mask is computed from the potential of the masked map (the nonlocal part of the gradient is missing near mask edges), which the per-band comparison with the convergence template bounds at a few per cent.
- *Magnification and selection.* The sightline quasars are magnified by the foreground; the mean field absorbs the resulting modulation of the pair density, as in the mocks, where the sightline draw carries the magnification.
- *Cross-correlation kernel.* The cross-correlation amplitude uses the kernel of the fixed-base fit; refitting with all parameters and the correction free (`report/stageb/dr1_qso.v1_free.json`) gives 0.64 ± 0.51 against 0.61 ± 0.52 , a +0.03 shift, 5 per cent of the amplitude’s error. The injection expectation has not been run for the quasar–pixel pairs, and the mocks contain no quasar–forest cross-correlation, so the cross-correlation estimator is validated by its nulls, by the consistency of the redshift and band splits with the auto-correlation, and by the QA of the fit only.
- *Not yet done.* The mock validation has not been repeated with region B, with the evolving correlation table or with the enlarged tracer set: the $A \times B$ machinery and the layered table are covered by unit tests (`tests/test_regions.py`, `tests/test_zevol.py`) and by the internal consistency of the runs, not by an end-to-end mock. Tomographic sub-slabs of the forest, the propagation of the bias uncertainties into the error, the joint slice fit, the 10 per cent residual amplitude difference between the two pair types at fixed redshift, the mock validation of the 1.96–2.1 extension (the mock box starts at $z = 2.1$), and an injection test and mock for the quasar–forest cross-correlation.

8 Conclusions

A pair-based quadratic estimator of the forest’s lensing deflection, projected on a template from DESI and BOSS galaxies and quasars at $0.1 < z < 1.6$, is unbiased on 400 self-lensed mocks with lognormal tracers and gives $A = 0.35 \pm 0.47$ on every DR1 forest pixel below $z = 3$ (effective redshift 2.35), using both delta regions, bands to $L = 500$, a redshift-evolving response kernel and eleven tracer samples, with clean curl, random-template and injection tests; the quasar–forest

cross-correlation, treated with the same estimator and an independent correlation model, gives $A = 0.61 \pm 0.52$, nearly uncorrelated with the auto-correlation, and the combination $A = 0.47 \pm 0.35$ (0.58 ± 0.31 with the bands extended to $L = 1000$). The template-side biases are measured internally with NaMaster, and the deflection templates are validated against ACT and Planck lensing, where they come out at 0.99 ± 0.04 (ACT) and 0.92 ± 0.03 (Planck) of their nominal normalisation, the two maps agreeing on common sky. The combined precision corresponds to a 1.5σ deficit from the Λ CDM value, better than forecast for DR1; the same pipeline on the complete DESI sample (roughly twice the forests and deeper coverage at $z > 2.5$) should reach $2\text{--}3\sigma$.

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